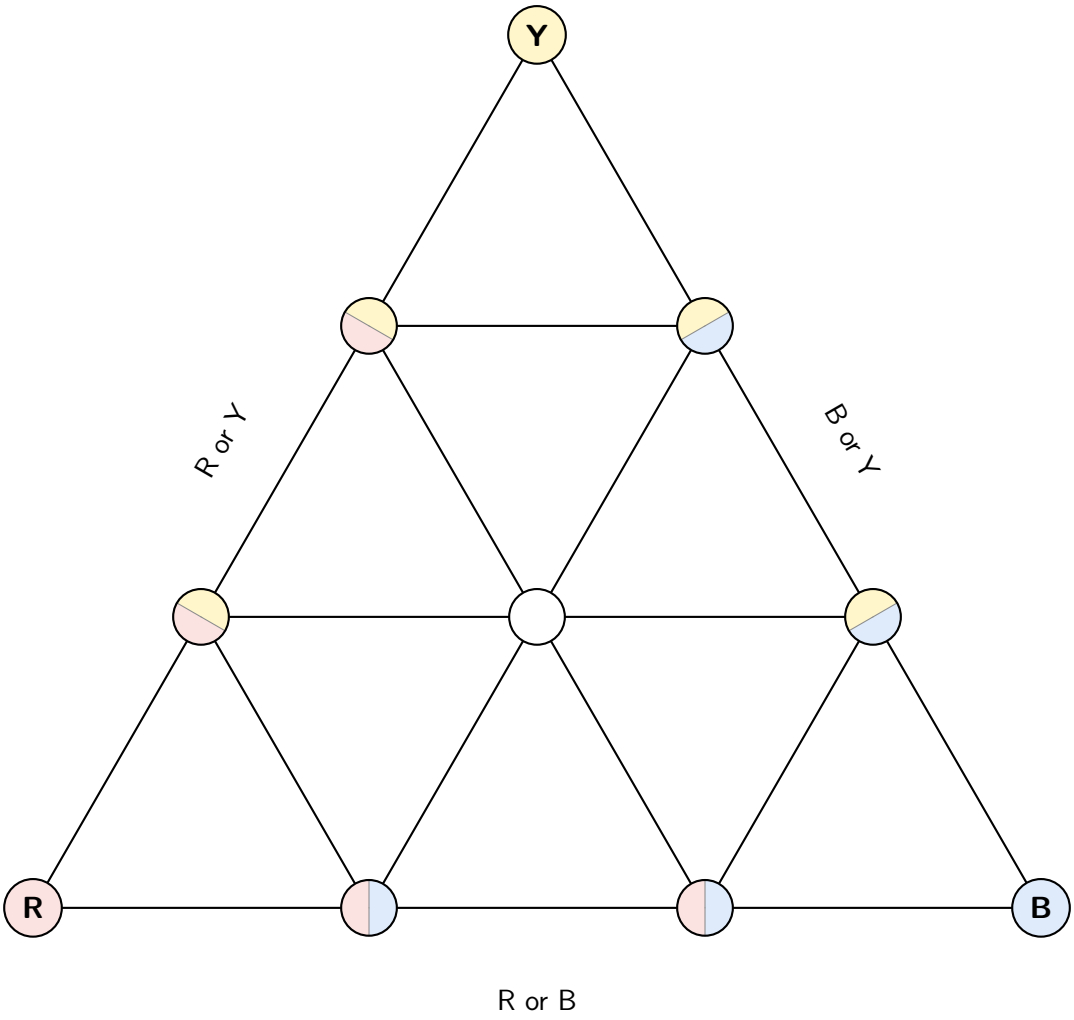
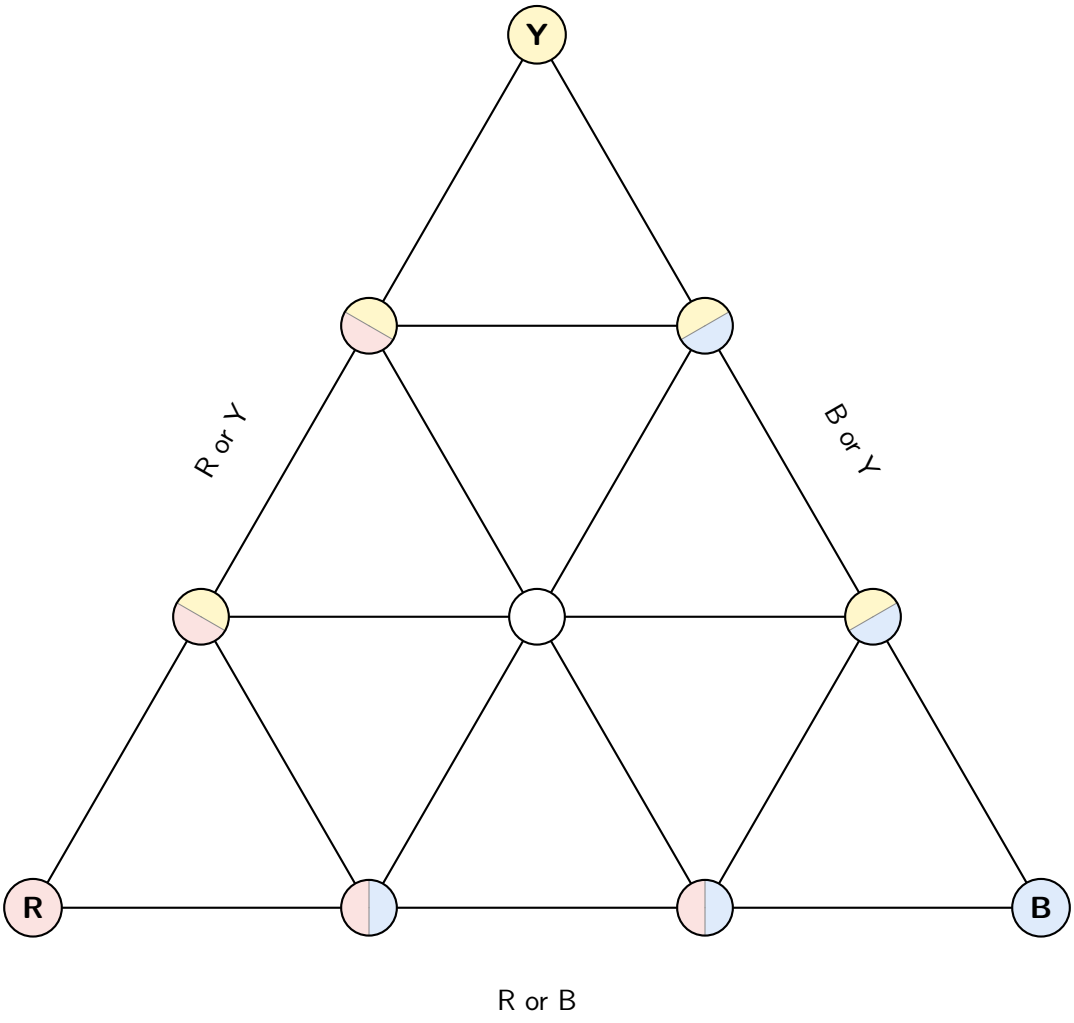
Write R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

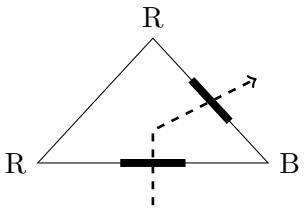
**Problem 1:** Find a filling with as few little triangles containing R, B, and Y as you can, and another with as many as you can. Which counts between your two records can you make?



Another board for Problem 1.



**Problem 2:** Call an edge with R at one end and B at the other a door. Mark every door. A route crosses through doors and uses no door twice. In a two-door triangle it must use both doors. Draw all the routes. Can every route from outside return outside?




Example: enter one thick door and leave through the other.
The dashed route crosses edges; it does not follow them.

